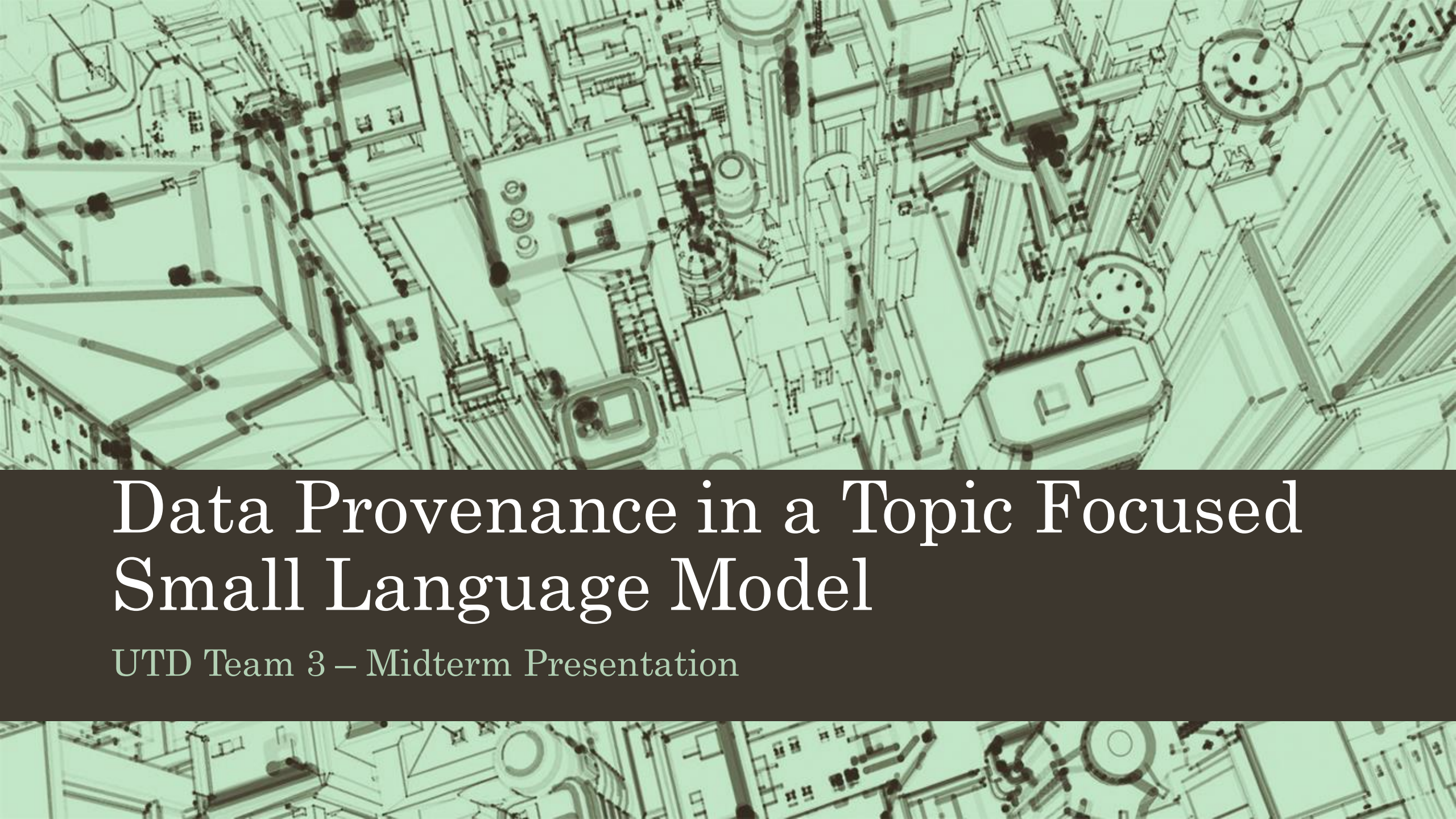




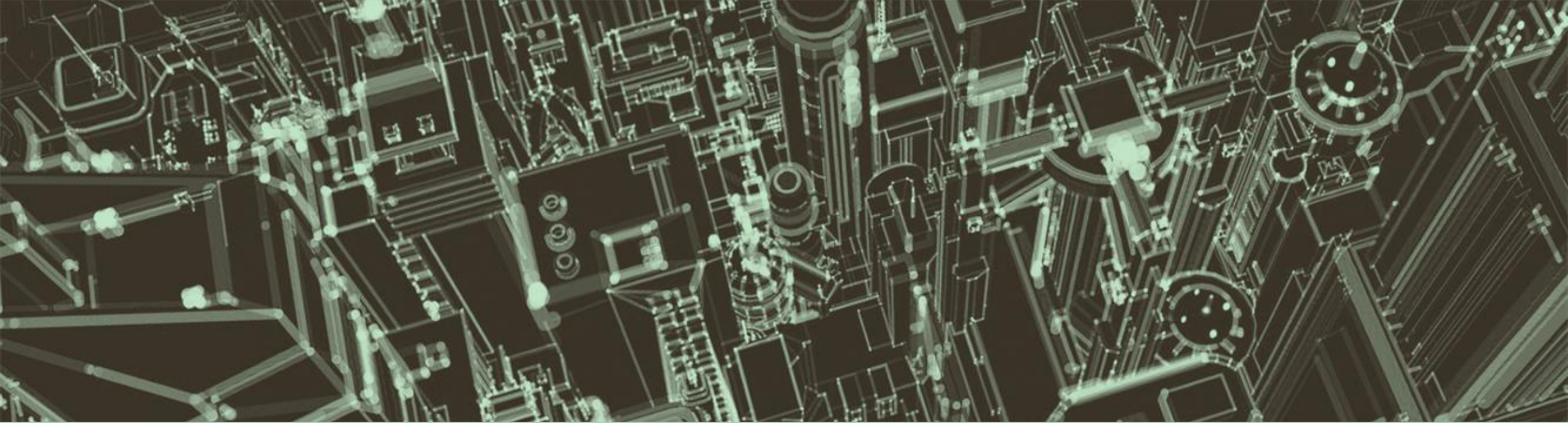
Data Provenance in a Topic Focused Small Language Model

UTD Team 3 – Midterm Presentation

MEET THE TEAM

The University of Texas at Dallas – TEAM 3

- Dhyana Sai Reddy Dalli (DXD210031)
- Subash Nirmal Kolluru (S XK210036)
- Aaryavardhun Guntupally (AXG210117)
- Yaswanth Kulas Devarasetty (YXD210007)
- MARK BOUTWEL (National Security Agency(NSA))
- BRIAN RICKS –FACULTY ADVISOR

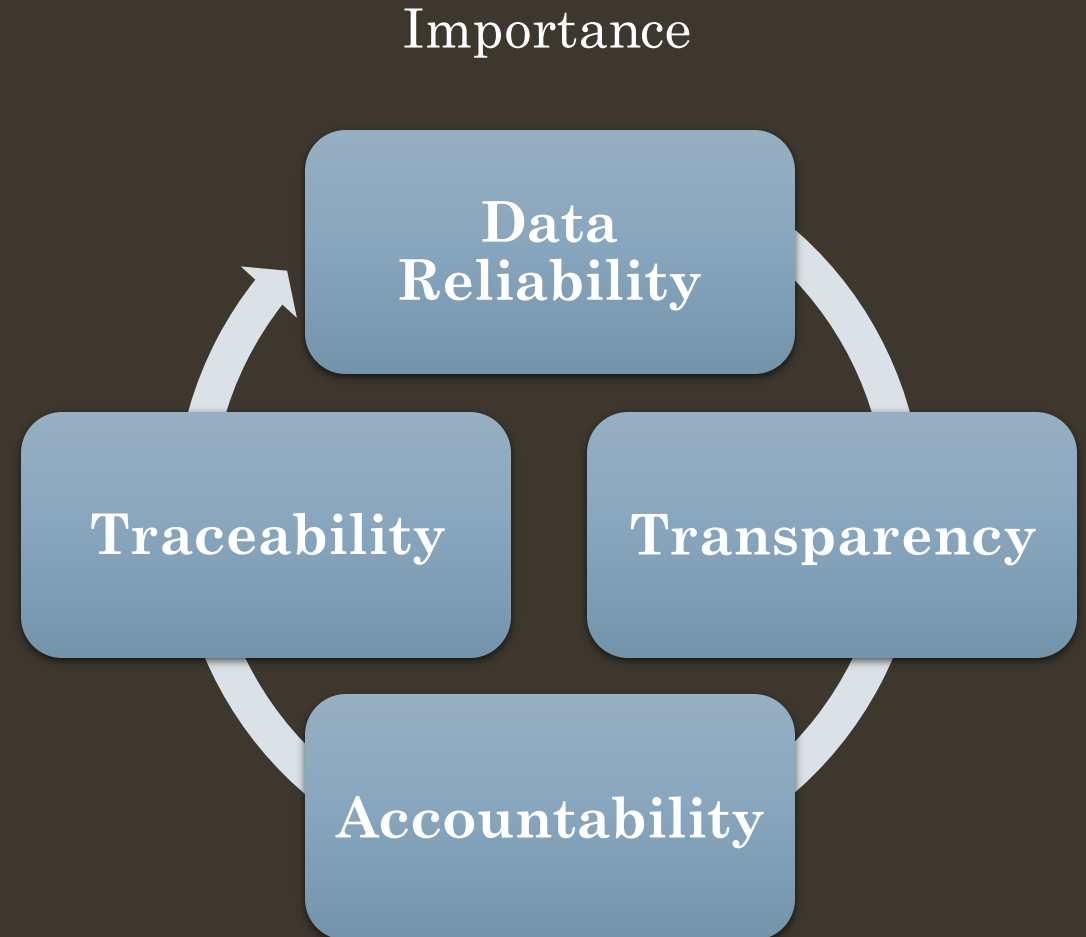


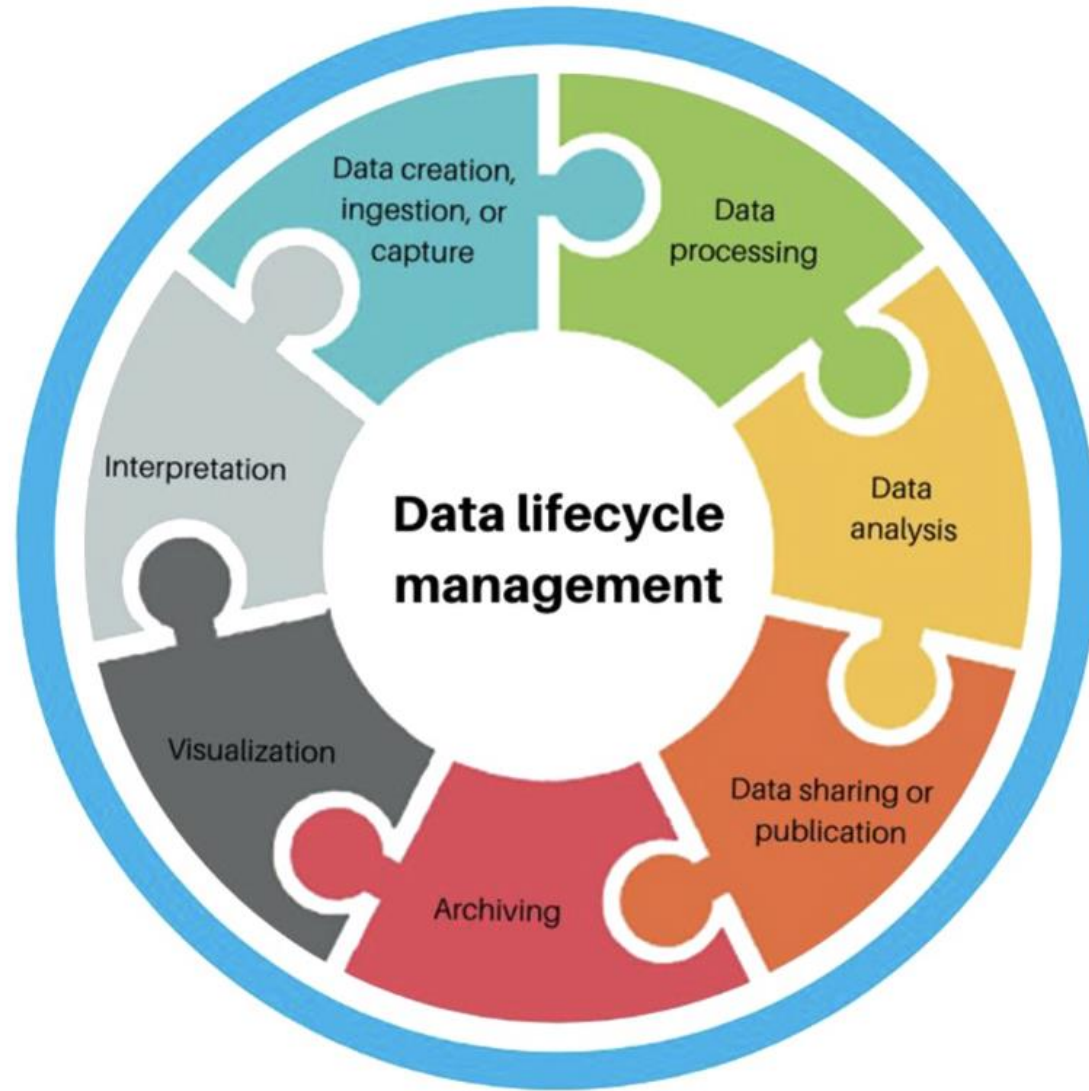
INTRODUCTION



INTRODUCTION – Data Provenance

- Definition: Data provenance, also known as data lineage, is the comprehensive documentation of the origin, history, and transformations applied to data throughout its lifecycle.





DATA LIFE CYCLE

Key Objectives

Project Focus: The primary focus of our project is to explore and enhance data provenance in Topic Focused Small Language Models (TFLMs).

A large, light blue downward-pointing arrow indicating a flow from the Project Focus section to the Benefits section.

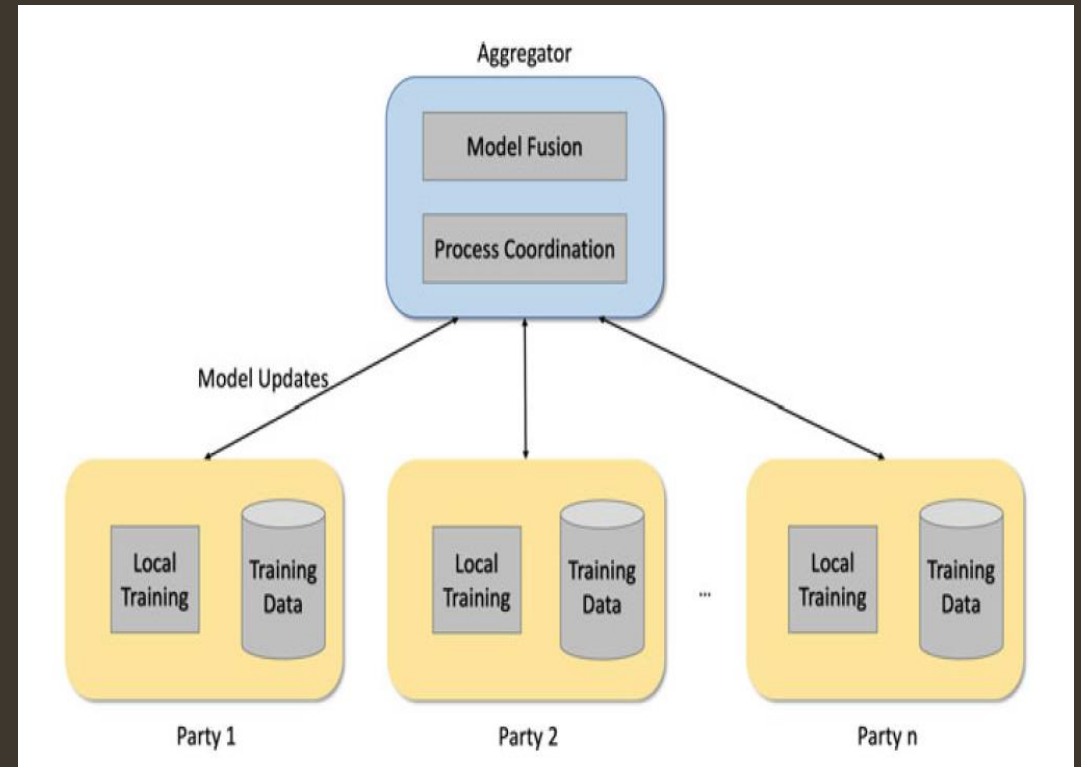
Benefits: By improving data provenance in TFLMs, we aim to make AI systems more transparent, accountable, and reliable.

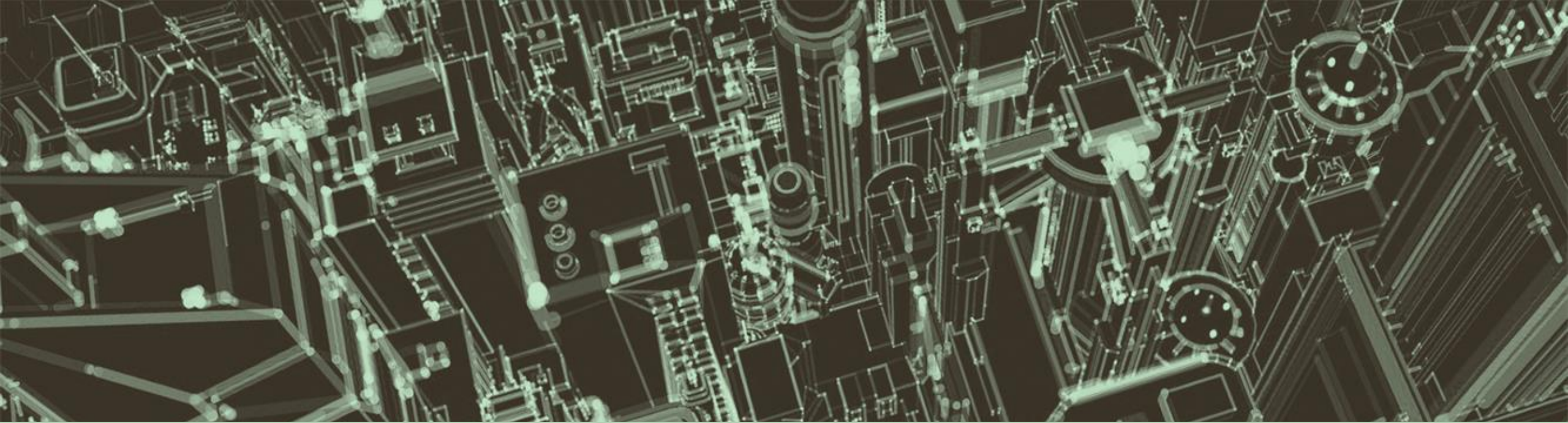
A large, light blue downward-pointing arrow indicating a flow from the Benefits section to the Applications section.

Applications: This initiative has broad applications in various industries, including healthcare, finance, and research, where data traceability and ethical considerations are of utmost importance.

ISSUES

- Privacy Concerns
- Unwanted Data Collection
- Model Inversion Attacks
- Data Poisoning





LITERATURE REVIEW & CONCEPTS

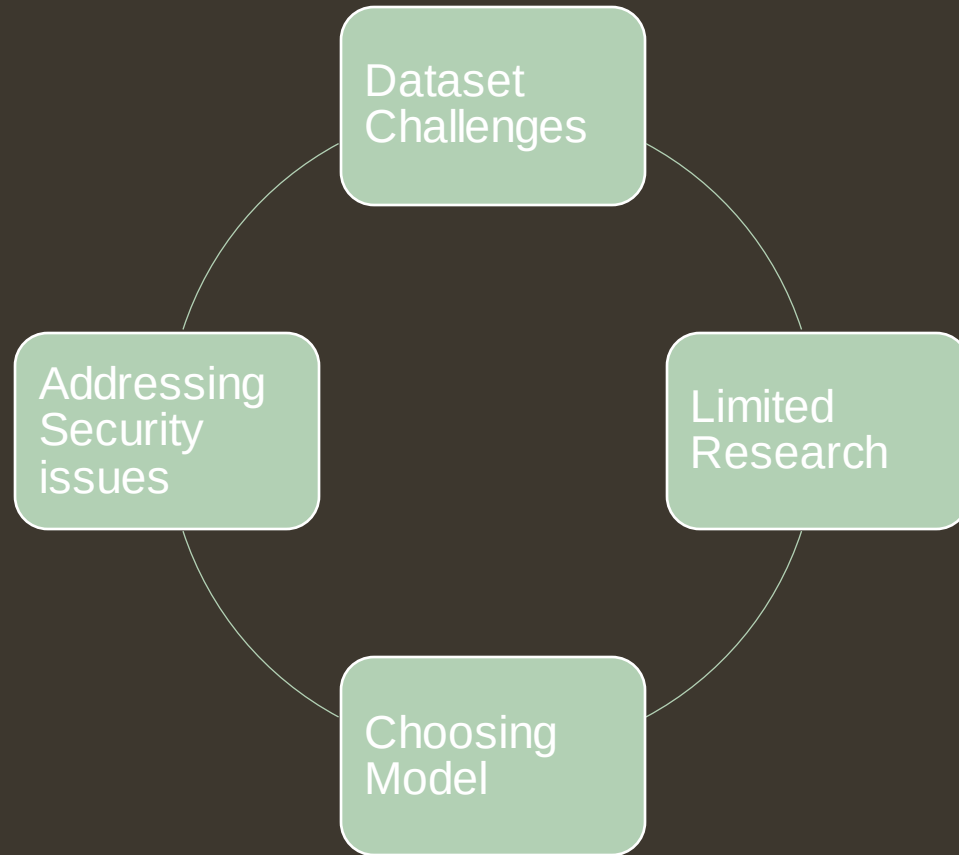


Key Research Papers

1. "SMILES Pair Encoding (SPE)" - Li & Fourches (2021)
 - Novel tokenization method for cheminformatics.
 - Enhances molecular generation and QSAR prediction that outperforms atom-level tokenization.
2. "Deep Learning Methods for Molecular Representation" - Li et al. (2022)
 - Applies deep learning to molecular representation and property prediction.
 - Discusses models and interpretability.
3. "EdnaML: Framework for Reproducible Deep Learning" - Suprem et al. (2022)
 - Introduces EdnaML for reproducible deep learning.
 - Offers automation and customization for ML pipelines.

These papers offer insights into techniques and frameworks applicable to enhancing data traceability in TFLMs.

Challenges



SAFETY OF MODELS

- The malicious attacker can completely steal the training data from gradients in a few iterations
- TFLM model designs may contain vulnerabilities for:
 1. The (potentially malicious) server
 2. Any Participant
- Insider vs Outsider
- Training Phase vs Inference Phase
- Poisoning Attacks

Training Phase vs Inference Phase

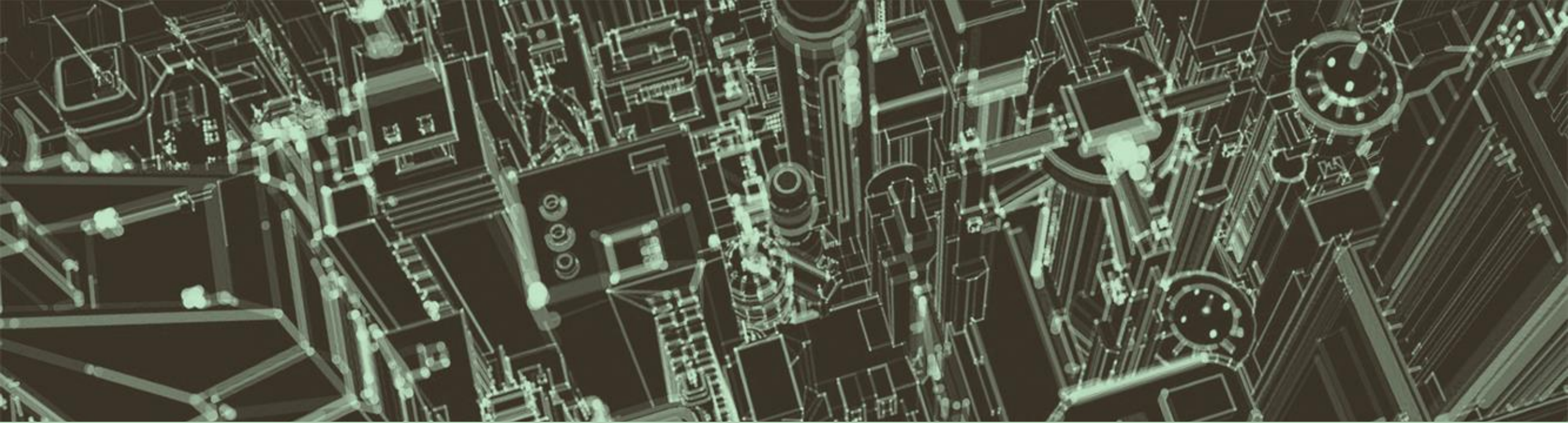
- Attacks at training phase attempt to learn, influence, or corrupt the TFLM model itself.
- During training phase, the attacker can run Data Poisoning Attacks or Model Poisoning Attacks.
- Attacks at inference phase do not tamper with the targeted model, but instead, either cause it to produce wrong outputs (targeted/untargeted) or collect evidence about the model characteristics.
- Inference phase attacks can be classified into:
 - White-box attacks (i.e., with full access to the TFLM model)
 - Black-box attacks (i.e., only able to query the TFLM model)

Poisoning Attacks

- Poisoning attacks can be:
 - a) Random attacks
 - b) Targeted attacks.
- Random attacks aim to reduce the accuracy of the ML model, whereas targeted attacks aim to induce the ML model to output the target label specified by the adversary.
- Poisoning attacks during the training phase can be performed on the data or on the model.
- Two poisoning attacks:
 - 1) Data poisoning (during local data collection).
 - 2) Model poisoning (during local model training process).

Data Poisoning

- Data poisoning is a cyber-attack that manipulates machine learning models by inserting false or malicious data.
- A malicious attack where an adversary intentionally injects incorrect or manipulated data into the training process of a federated learning system.
- This can cause the model to make incorrect or biased predictions, with potentially serious consequences in critical applications like Medical Diagnosis or Finance.
- Data poisoning attacks largely fall in two categories:
 - 1) Clean-Label
 - 2) Dirty-Label
- One common example of dirty-label poisoning attack is the Label-Flipping attack.
- Backdoor Poisoning.



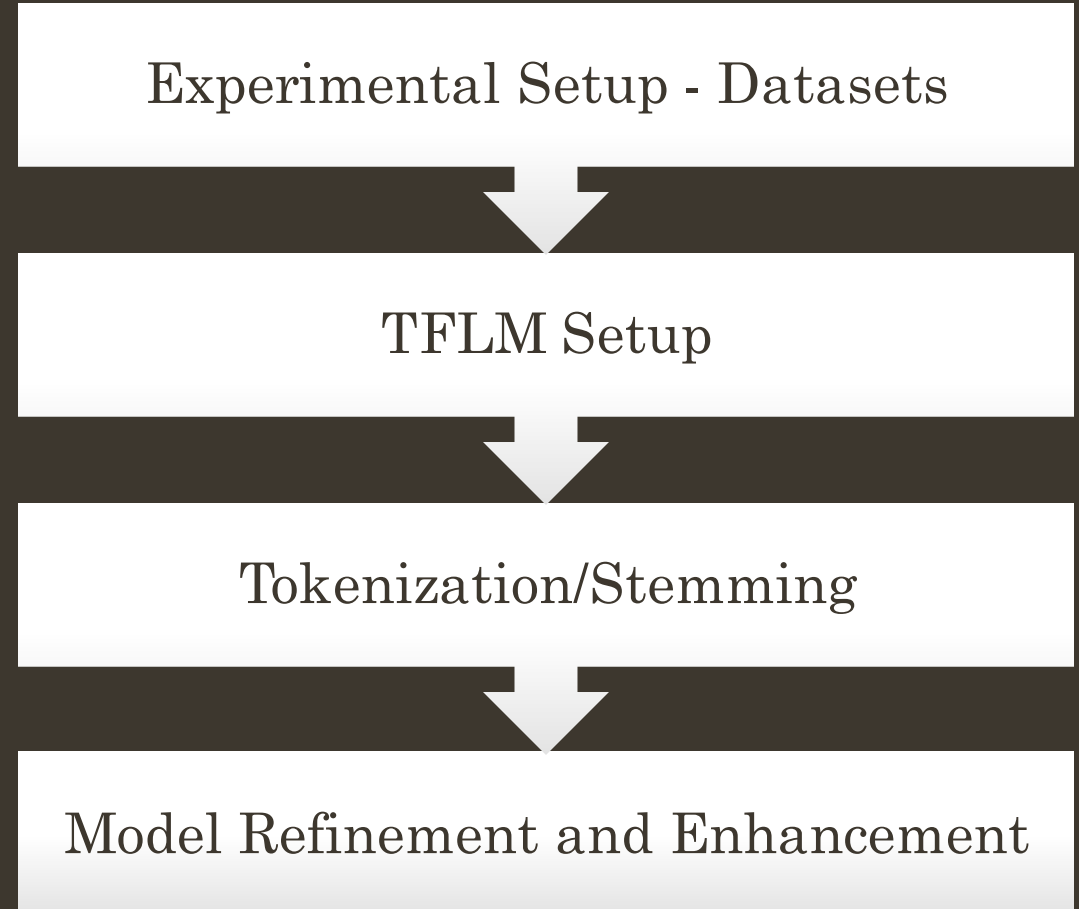
NEXT STEPS

What difference can we make?



ANALYZING STUDIES & CONDUCTING PILOT REPRODUCTIONS

- Work on various datasets and focusing on their frameworks, methodologies and data provenance techniques.
- Data and Framework Reproduction and Customization
- Compare the findings from the first set of pilots (original frameworks and datasets) and the second set of pilots (customized frameworks and datasets).
- Train the model based on the findings from the second pilot and change them based on the accuracy obtained.



DEFENDING AGAINST LABEL FLIPPING ATTACKS

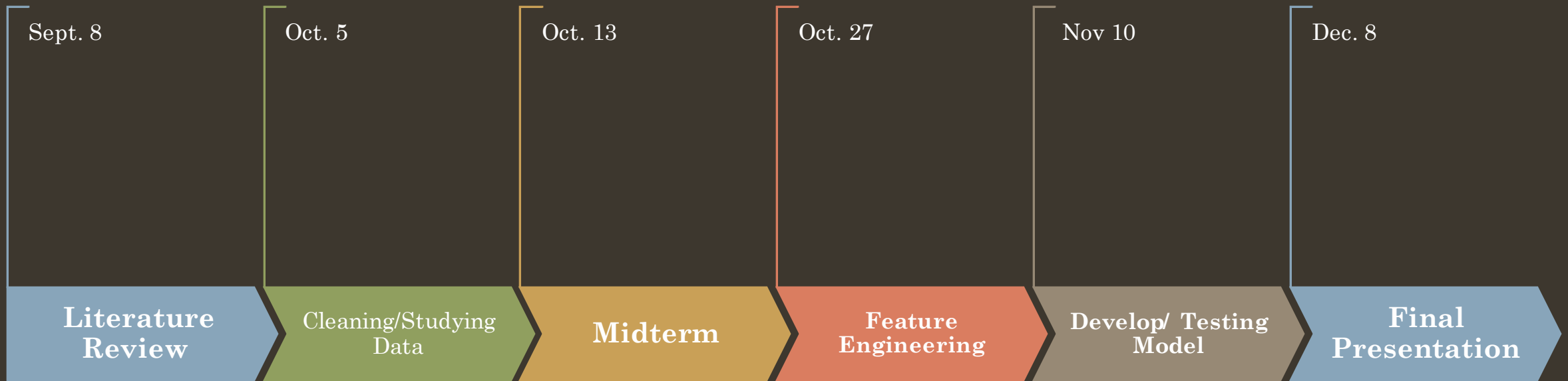
- Algorithms to identify Malicious model Updates in Models.
- Improve the existing Algorithms for a better efficiency.

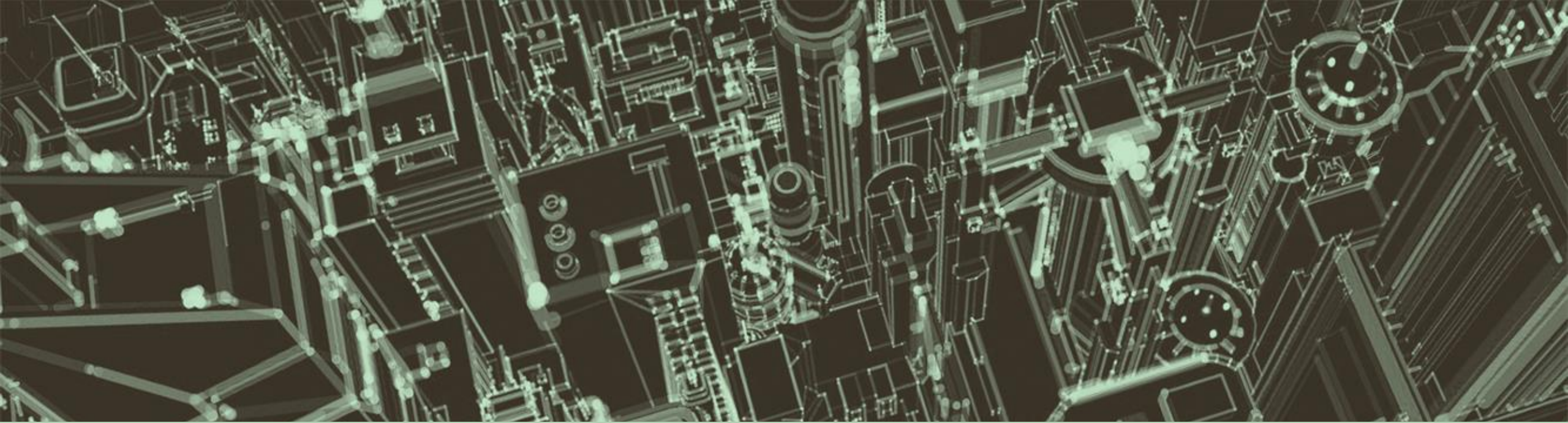
Dimensionality
Reduction
(TF-IDF)



Improve Existing
Algorithms

TIMELINE





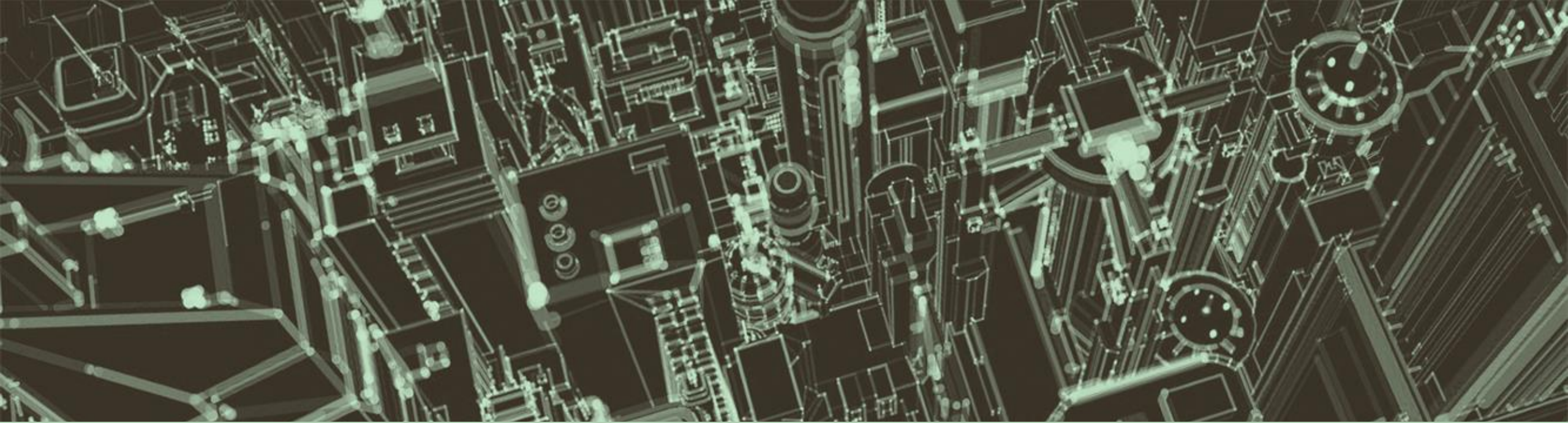
CONCLUSION

SUMMARY



SUMMARY

1. **Comprehensive Data Provenance Exploration.**
2. **Strategic Approach with Small Targeted Datasets.**
3. **Impactful Contributions and Responsible AI Development.**



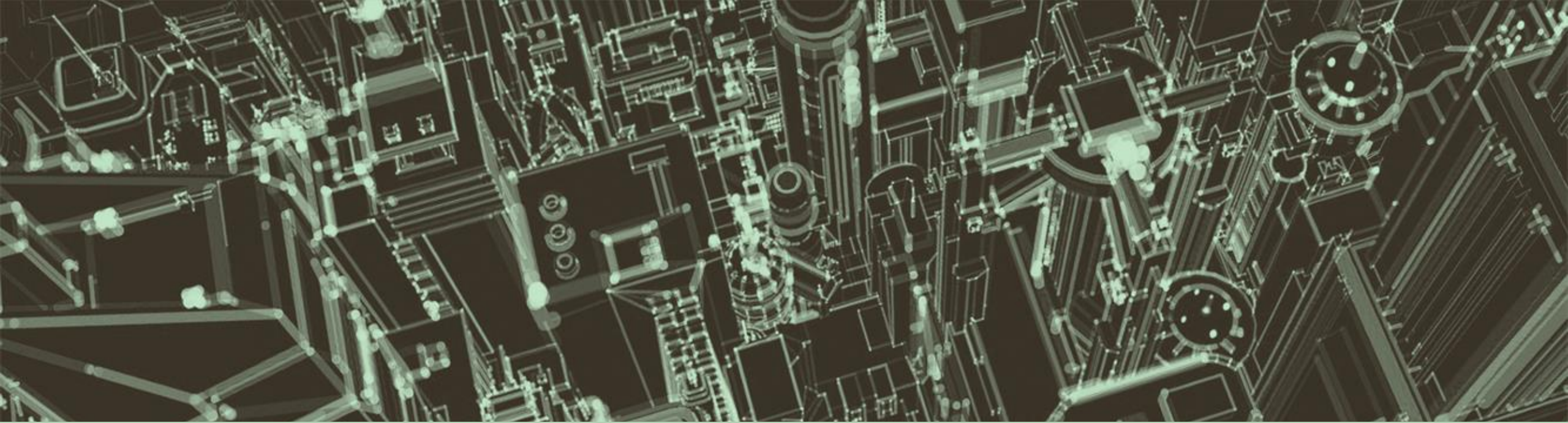
REFERENCES

All the Research papers read and referenced.



REFERENCES

1. Li, X., & Fourches, D. (2021). SMILES pair encoding: a data-driven substructure tokenization algorithm for deep learning. *Journal of chemical information and modeling*, 61(4), 1560-1569.
2. Li, Z., Jiang, M., Wang, S., & Zhang, S. (2022). Deep learning methods for molecular representation and property prediction. *Drug Discovery Today*, 103373.
3. Longpre, S., Mahari, R., Muennighoff, N., Chen, A., Perisetla, K., Brannon, W., ... & Hooker, S. The Data Provenance Project.
4. Suprem, A., Vaidya, S., Venugopal, A., Ferreira, J. E., & Pu, C. (2022). EdnaML: A Declarative API and Framework for Reproducible Deep Learning. *arXiv preprint arXiv:2211.06783*.



THANK YOU

Any Questions ?

